GLOBAL AGRO-PROCESSING BLUEPRINT

Ahmedabad Premium Dehydrated Food Export Hub (APFE)

A detailed business proposal for establishing a high-efficiency food dehydration and processing plant in Bavla GIDC, targeting premium export markets in Europe and North America.

PROJECT PROMOTER

Lead Entrepreneurial Group

PRIMARY FACILITY ZONE

Bavla GIDC (Ahmedabad Rural)

PHASE 1 CAPITAL PLAN

₹25 Lakhs (Shed Lease & Dryer Setup)

TARGET MARKET

EU, US, and Middle East B2B Retail

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1. Executive Summary

India is the second-largest producer of onions and garlic globally. However, post-harvest losses due to storage moisture and price volatility regularly impact farmers. Dehydrating fresh agricultural products into flakes and powders reduces weight by up to 80% and extends shelf life to over a year, creating a viable export commodity.

The proposed **Ahmedabad Premium Dehydrated Food Export Hub (APFE)** is an agro-processing facility to be established in Bavla GIDC. With a Phase 1 capital of ₹25 Lakhs, the plant will use continuous belt hot-air steam dryers to process fresh onions and garlic sourced from local mandis, converting them into premium dehydrated products for export.

APFE will target premium food manufacturing and retail packaging companies in Europe and North America. The facility will implement strict quality controls, testing all batches for pesticide residues and heavy metals to ensure compliance with export market regulations.

Core Business Intent

By establishing a compliant food processing unit in the agricultural belt of Bavla, APFE connects local raw material sourcing with high-margin international B2B buyer networks.

2. Future Potential of the Industry

The global dehydrated food market is projected to reach $18B by 2030, driven by the growth of packaged food sectors, instant noodle manufacturers, and B2B spice blenders.

By processing fresh vegetables during peak harvest seasons when prices are low, dehydration units lock in high margin spreads for export sales during off-season periods.

2.1 FDA & EU Pesticide MRL Regulations

To enter premium export markets, products must meet strict food safety standards:

Import Jurisdiction	Pesticide MRL Rules (Regulation 396/2005)	Fumigant Bans	Mandatory Documentation
European Union (EU)	Default limit of **0.01 mg/kg** for unauthorized compounds. Dehydration factors allowed (~3x fresh limit).	**Ethylene Oxide (ETO) completely banned.** Consignments containing ETO are rejected.	Certificate of Analysis (COA) from an NABL-accredited laboratory; APEDA/AGMARK clearance.
United States (US FDA)	Zero-tolerance for unapproved chemical residues. Compliance with FSMA rules.	ETO allowed under strict limits, but chemical-free heat treatment preferred.	FDA Food Facility Registration Number; Prior Notice filings for all shipments.

2.2 Dholera Logistics Corridor & Olympics 2036 Demand

Logistics and local market events support the project's long-term business case:

- **Dholera Cargo Terminal Access:** The construction of the **Dholera International Airport** and its dedicated cargo hub simplifies air freight logistics to Europe and the Middle East, reducing transit times from Ahmedabad factories.
- **Ahmedabad Olympics 2036:** The bid for the 2036 Olympics will drive demand for convenient, high-shelf-life food options for athletes and visitors. APFE can supply dehydrated ingredients to institutional catering networks.

3. Location & Mandi Sourcing Strategy

Siting the facility in an agricultural GIDC zone reduces raw material transport costs:

Proposed Siting Zone	Suitability Rating	Raw Material Logistics Pros	Infrastructure Cons & Overheads	Verdict
Bavla GIDC (Ahmedabad Outskirts)	**9.5 / 10** 🏆 Recommended Siting	• Located directly on the Ahmedabad-Rajkot national highway. • Close to major onion and garlic mandis in Mahuva and Gondal. • Lower local land and labor overheads.	• Further from the Mundra port cargo terminals (~320 km).	**Winner for Phase 1.** Optimal balance between raw material proximity and logistics.
Sanand GIDC (Industrial Zone)	**7.0 / 10** ⭐ ⭐ Average Siting	• Stable power grid and clean water access. • Close to major industrial units.	• Higher lease premiums due to competition from automotive units.	**Too expensive** for agricultural processing operations.

4. Competitor Analysis

The agricultural dehydration market includes large processors and unorganized units:

- **Mahuva Dehydration Hubs (Large Local Clusters):**
  Model: Run high-volume solar drying yards producing low-cost dehydrated onion slices for the domestic wholesale market.
  APFE Counter-Strategy: We avoid the domestic commodity market and focus entirely on high-grade steam-dried powders certified for export markets.
- **Jain Irrigation (Dehydrated Foods Division):**
  Model: Operate large contract farming setups producing high-end dehydrated vegetables for international food brands.
  APFE Counter-Strategy: We focus on smaller, flexible B2B private-label shipments, offering faster delivery timelines to medium-sized overseas buyers.

5. Legal Compliance Matrix

Clearance Requirement	Issuing Authority	Est. Setup Cost	Timeline	Role in Export Lifecycle
FSSAI Central Export License	FSSAI Delhi Office	₹15,000	1 Month	Mandatory to legally pack, process, and export food items from India.
APEDA Registration (RCMC)	APEDA Ministry of Commerce	₹10,000	1 Month	Required to export agricultural products and participate in trade exhibitions.
US FDA Facility Registration	US Department of Health	₹25,000	2 Months	Required for all facility imports into the United States market.
Phytosanitary Certification	Plant Quarantine Dept	₹5,000 (per batch)	1 Week	Certifies that shipments are free from pests and plant diseases.

6. Raw Material Sourcing Channels

Sourcing raw materials at competitive rates is key to maintaining margins:

1. **Mahuva Onion Mandi (Bhavnagar):** Source fresh white onions directly from the largest onion mandi in Gujarat during harvest peaks.
2. **Gondal Garlic Mandi (Rajkot):** Sourcing fresh garlic bulbs during harvest cycles to secure optimal material yields.
3. **Contract Farmer Ties (Dholka Rural):** Establish direct supply agreements with farmers in nearby villages, providing them with quality seeds in exchange for guaranteed harvest buy-back rates.

7. Buyer & Broker Distribution Channels

We distribute processed products through established export channels:

- **International B2B Spice Brokers:** Partner with brokers in Rotterdam and London who distribute ingredients to European food factories.
- **Private Label Packers:** Supply premium dehydrated powders to packaging firms selling organic spices under retail brands in North America.
- **Institutional Food Manufacturers:** Form direct supply contracts with local and international noodle and soup mix manufacturers.

1

Shed Lease & Prep

Lease an industrial shed in Bavla GIDC and seal the floors with food-grade epoxy coating.

2

Dryer Installation

Install our continuous belt hot-air steam dryers and connect the steam piping.

3

FSSAI & APEDA Sync

Complete central export registrations and secure clearance for our processing facility.

4

Lab Testing Setup

Partner with an NABL-accredited laboratory to handle pesticide residue testing for all export batches.

9. Financial Projections

Phase 1 budget focuses on shed leasing, dryer setup, and initial raw material sourcing:

CapEx Budget Component	Cost (INR)	Monthly Operational Costs	Cost (INR)
Continuous Belt Steam Dryer	₹12,00,000	**Raw Agricultural Sourcing**	₹2,50,000
Food-Grade Epoxy Flooring	₹2,00,000	**Boiler Fuel (Biomass Pellets)**	₹40,000
APEDA & US FDA Registrations	₹1,50,000	**Labor Payroll (4 workers)**	₹80,000
Stainless Steel Washing Tanks	₹2,50,000	**Power Utility & Water Rates**	₹30,000
Working Capital Reserves	₹7,00,000	**Shed Rental (Bavla GIDC)**	₹30,000
Total CapEx Setup	**₹25,00,000**	**Total Monthly Burn**	**₹4,30,000**

Export Profit Margins

Processing 15 Tonnes of fresh onions yields 3 Tonnes of dehydrated product. Sourced at ₹15/kg and exported at ₹220/kg, this returns a gross margin of **₹6,60,000** per batch, covering operational costs and showing a payback period of under **15 months**.

10. Extended Growth Model

We plan to scale our processing capacity and expand our product range over a three-stage roadmap:

- **Phase 1: Basic Flakes & Powders (Year 1):** Focus on exporting dehydrated onion and garlic flakes to B2B blenders.
- **Phase 2: Specialty Powders & Blends (Year 3):** Install grinding mills to produce specialized spice mixes and custom culinary powders.
- **Phase 3: Retail Brand Launch (Year 5):** Launch our own brand of organic spices, targeting direct consumer sales in international markets.

11. Risk Analysis & Mitigation

Identified Business Risk	Impact Level	Defensive Mitigation Strategy
Pesticide Limit Violations	**High**	Test raw material batches for pesticide residues before processing, rejecting low-quality lots.
Crop Price Spikes	**Medium**	Purchase and store raw material reserves during peak harvest seasons when market prices are lowest.
Moisture Damage in Transit	**Medium**	Pack finished products in multi-layer vacuum-sealed bags to prevent moisture absorption.

11.1. Worst-Case Stress-Testing & Liquidation Playbook

Stress Scenario	Immediate Financial Impact	Pre-Planned Mitigation & Escape Route
EU customs rejects a shipment due to chemical residues	Loss of ₹12 Lakhs cargo value; return shipping costs of ₹2 Lakhs.	Sell the rejected shipment to local domestic starch and spice manufacturers to recover costs.
Liquidation Trigger (Stop-Loss)	Total accumulated losses exceed ₹10 Lakhs or raw material costs spike by 80% for 6 consecutive months.	**Capital Preservation Liquidation Protocol:** 1. Halt all raw agricultural sourcing and clear out active processing batches. 2. Sell our drying machinery and steel tanks to local processors to recover capital. 3. Cancel our facility leases and return unused capital to investors.

12. Steam Dehydration Technology

Our processing facility uses multi-stage conveyor belt hot-air dryers to maintain product quality.

The drying process uses steam heat exchangers to maintain an drying chamber temperature of **55°C to 65°C**. Slicing machines process fresh onions to a thickness of **3mm to 5mm** to optimize drying times.

Our steam boilers use clean biomass briquettes, maintaining a steam pressure of **6 to 8 kg/cm2** to support continuous drying operations.

13. HACCP Hygiene & Safe EHS Protocols

To maintain hygiene standards, all workers must wear hairnets, gloves, and protective boots.

Our food safety protocols are structured to prevent biological contamination:

- **Microbial Controls:** Test processed batches to ensure that *Salmonella* and *E. coli* are completely absent before packaging.
- **HACCP Compliance:** Install metal detectors on our packaging lines to prevent contamination in finished bags.

14. Farmer Cooperatives & Local Labor

We source our sorting and processing labor from nearby communities in Bavla. APFE partners with local farmer cooperatives to secure a reliable supply of raw materials and support regional employment.

15. Strategic Summary & Verdict

🏆 Investment Verdict: GO (Highly Recommended)

Establishing the **Ahmedabad Premium Dehydrated Food Export Hub (APFE)** is highly recommended. The stable export demand for dehydrated products and our proximity to key agricultural mandis provide a strong foundation for high-margin operations.

Extended Go-Scale Target Model

Growth Phase	Processing Vol. (T/year)	Average Margin	Projected Revenues
Phase 1: Flakes & Powders	180 Tonnes	18% - 22%	₹1.8 Crores
Phase 2: Specialty Blends	600 Tonnes	24% - 28%	₹7 Crores
Phase 3: Retail Brand	2,000 Tonnes	30% - 35%	**₹28 Crores**

Liquidation & Risk Safeguards

- **Pre-shipment audits:** Verify the quality of all batches before exporting to prevent customs rejections.
- **Seasonal sourcing:** Purchase raw materials during harvest peaks to minimize raw material cost impacts.